

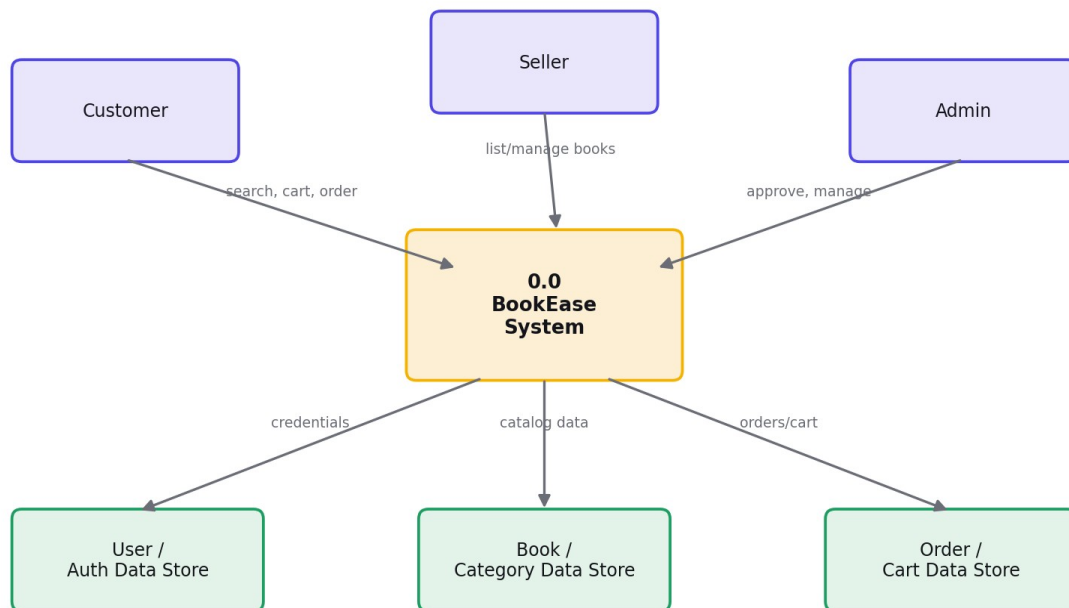
Project Design Phase-II

Data Flow Diagram & User Stories

Date	19 July 2026
Team ID	[Enter Your Team ID]
Project Name	BookEase – Multi-Role Online Bookstore Platform
Maximum Marks	4 Marks

Data Flow Diagram — Level 0 (Context Diagram)

The Level-0 DFD shows BookEase as a single process exchanging data with its three external entities (Customer, Seller, Admin) and its three core data stores.



Data Flow Diagram — Level 1 (Decomposition)

The system process decomposes into four sub-processes:

- 1.1 Authentication Process — validates credentials, issues JWTs, and reads/writes the User data store for all three roles.
- 1.2 Catalog Process — handles book/category browsing, search and seller-submitted listings, reading/writing the Book/Category data store.
- 1.3 Cart & Order Process — manages cart operations and order placement/cancellation, updating stock in the Book store and writing to the Order/Cart data store.
- 1.4 Admin Process — manages seller approvals, user status/ratings, and cross-cutting oversight of books and orders.

User Stories

The table below lists the user stories driving BookEase's implementation, grouped by user type.

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Customer	Registration	USN-1	As a customer, I can register by entering my name, email, mobile and password.	Account is created and I am automatically logged in.	High	Sprint-1
Customer	Login	USN-2	As a customer, I can log in using my email and password.	A JWT is issued and I am redirected to the home page.	High	Sprint-1
Customer	Profile	USN-3	As a customer, I can update my profile details and upload a profile picture.	Changes are saved and reflected immediately on the profile page.	Medium	Sprint-1
Customer	Book Browsing	USN-4	As a customer, I can search and filter books by genre, author, price and rating.	Matching results are shown with pagination.	High	Sprint-1
Customer	Wishlist	USN-5	As a customer, I can save books to a wishlist for later.	Saved books appear on my Wishlist page.	Medium	Sprint-2
Customer	Cart	USN-6	As a customer, I can add, update and remove items in my cart.	The cart total recalculates instantly and persists across login sessions.	High	Sprint-2
Customer	Checkout	USN-7	As a customer, I can place an order with a shipping address and payment method.	Order is created, stock is decremented, and the cart is cleared.	High	Sprint-2
Customer	Orders	USN-8	As a customer, I can view my orders and cancel one by providing a reason.	Cancelled order stores my reason and restocks the book automatically.	High	Sprint-3
Seller	Registration	USN-9	As a seller, I can register with my business name, email and password.	Account is created with a "pending" approval status.	High	Sprint-3
Seller	Book Listing	USN-10	As a seller, I can add, edit and delete my own book listings.	Only books I own can be edited or deleted by me.	High	Sprint-3
Seller	Order Fulfillment	USN-11	As a seller, I can view and update the status of orders containing my	Status changes are reflected on the customer's order page.	Medium	Sprint-4

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
			books.			
Administrator	Seller Management	USN-12	As an admin, I can approve or reject new seller accounts.	Only approved sellers can list books.	High	Sprint-3
Administrator	User Management	USN-13	As an admin, I can activate, suspend or delete users and set their rating.	Status/rating changes take effect on the user's next request.	Medium	Sprint-4
Administrator	Book & Category Management	USN-14	As an admin, I can add, update and remove books and categories from any seller.	Catalog updates are visible to customers immediately.	High	Sprint-1
Administrator	Order Management	USN-15	As an admin, I can view all orders and update their status.	The updated status is visible on the customer's order page.	High	Sprint-4